

HAIFA REFINERIES, Israel

WMO: 401550

Lat: 32.8034N

Lon: 35.0548E

Elev: 9

StdP: 101.22

Time Zone: 2.00 (ISR)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.4	8.6	-6.5	2.2	19.4	-3.0	2.9	17.7	11.1	16.7	9.8	15.6	2.4	150	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	33.0	21.4	31.8	23.1	31.0	23.2	25.6	29.8	25.2	29.5	24.7	29.2	4.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	19.3	28.5	23.8	18.6	28.3	23.2	18.0	28.1	79.2	30.0	77.1	29.7	75.3	29.2	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.9	6.9	DB	5.0	38.4	1.5	2.1	4.0	39.9	3.1	41.1	2.3	42.3	1.2	43.8	(4)
			WB	3.3	26.7	1.6	1.0	2.1	27.4	1.2	28.0	0.2	28.5	-0.9	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.8	13.9	14.0	15.7	18.8	21.9	25.1	27.5	27.9	26.4	23.4	19.5	15.3	(6)
(7)		DBStd	5.57	2.15	2.32	2.73	3.38	2.63	1.63	1.09	1.13	1.44	2.49	2.95	2.22	(7)
(8)		HDD10.0	3	1	1	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	505	139	122	92	29	2	0	0	0	0	1	21	98	(9)
(10)		CDD10.0	3952	121	114	177	263	369	453	544	555	491	417	284	164	(10)
(11)		CDD18.3	1413	1	2	11	43	113	203	286	297	241	159	54	3	(11)
(12)		CDH23.3	13104	1	6	69	316	541	1612	3207	3536	2449	1146	218	4	(12)
(13)	CDH26.7	3889	0	1	17	121	139	349	1095	1289	627	221	29	0	(13)	
(14)	Wind	WSAvg	3.2	3.5	3.4	3.3	3.4	3.2	3.0	2.9	2.9	3.0	2.9	3.2	3.5	(14)
(15)	Precipitation	PrecAvg	614	158	114	56	23	5	0	1	0	2	31	86	132	(15)
(16)		PrecMax	878	321	278	166	90	25	7	13	2	14	91	272	292	(16)
(17)		PrecMin	297	44	29	9	0	0	0	0	0	0	0	15	20	(17)
(18)		PrecStd	164	69	60	41	22	6	1	3	0	4	28	66	73	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	23.7	29.0	34.1	35.0	32.9	34.0	33.3	33.2	34.7	29.5	23.4	(19)
(20)		0.4%	MCWB	13.6	13.0	14.9	16.9	17.5	21.7	23.4	25.2	20.2	17.5	16.4	13.8	(20)
(21)		2%	DB	20.1	21.0	24.9	30.3	30.0	30.6	32.3	32.0	31.0	30.9	27.2	21.6	(21)
(22)		2%	MCWB	12.4	12.3	14.2	16.4	17.6	22.0	23.7	24.7	23.5	18.5	15.9	13.8	(22)
(23)		5%	DB	18.8	19.2	22.4	27.0	27.5	29.4	31.4	31.4	30.1	28.5	25.6	20.4	(23)
(24)		5%	MCWB	12.8	12.3	14.0	15.8	18.4	22.2	23.7	24.4	23.3	20.9	16.2	13.3	(24)
(25)		10%	DB	17.8	18.0	20.0	23.9	25.9	28.6	30.6	30.7	29.3	27.5	24.3	19.3	(25)
(26)		10%	MCWB	12.5	12.7	14.4	15.4	19.2	21.9	23.6	24.1	22.9	21.3	16.6	13.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	15.7	17.8	19.6	22.0	24.4	26.2	26.7	25.8	24.9	21.1	16.9	(27)
(28)		0.4%	MCDWB	19.2	19.4	22.7	27.2	25.7	28.0	30.2	30.8	29.4	27.9	25.0	20.4	(28)
(29)		2%	WB	14.6	14.7	16.7	18.2	21.3	23.7	25.3	25.8	25.1	23.7	19.8	15.7	(29)
(30)		2%	MCDWB	17.9	18.6	20.9	23.4	25.1	27.9	29.7	30.0	29.0	27.2	23.7	19.3	(30)
(31)		5%	WB	13.8	14.0	16.0	17.5	20.7	23.2	24.7	25.4	24.4	23.0	18.9	14.7	(31)
(32)		5%	MCDWB	17.4	17.6	19.6	22.5	24.7	27.5	29.4	29.8	28.6	26.8	23.3	18.4	(32)
(33)		10%	WB	13.1	13.2	15.2	16.9	20.2	22.7	24.4	25.0	23.9	22.2	17.9	14.0	(33)
(34)		10%	MCDWB	16.7	16.9	19.0	21.8	24.2	27.1	29.0	29.6	28.3	26.2	22.4	17.9	(34)
(35)	Mean Daily Temperature Range		MDBR	6.6	7.2	7.4	7.7	7.3	6.6	6.4	5.9	6.1	7.4	8.1	6.8	(35)
(36)		5% DB	MCDBR	8.2	9.2	11.3	13.6	10.9	8.1	7.7	6.6	7.1	9.2	9.7	8.4	(36)
(37)		5% DB	MCWBR	4.5	5.2	5.2	4.9	4.5	3.3	3.0	2.5	3.3	5.1	5.1	4.2	(37)
(38)		5% WB	MCDBR	7.1	7.8	8.1	8.8	8.0	7.4	6.9	6.1	6.9	7.1	8.1	7.1	(38)
(39)		5% WB	MCWBR	4.5	5.2	4.9	4.3	3.9	3.4	2.8	2.6	3.8	4.9	5.4	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.387	0.419	0.459	0.503	0.499	0.432	0.450	0.455	0.438	0.449	0.405	0.389	(40)	
(41)		taud	2.273	2.164	2.050	1.948	2.000	2.237	2.197	2.183	2.232	2.172	2.265	2.280	(41)	
(42)		Ebn at Noon	812	821	816	797	803	855	838	828	827	780	784	784	(42)	
(43)		Edh at Noon	109	134	161	186	178	140	145	145	133	131	109	103	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.81	3.65	5.09	6.28	7.38	8.11	7.88	7.17	6.11	4.58	3.37	2.62	(44)	
(45)		RadStd	0.20	0.32	0.31	0.34	0.31	0.14	0.09	0.13	0.15	0.21	0.18	0.20	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (17 neighbors)	+0.44	N/A	-1.92	N/A	N/A	N/A	N/A	-70	+160	+107	(47)

Nomenclature: See separate page